

Continuity and the Intermediate Value Theorem



The three-part continuity test

- 1 State the three conditions that must all hold for f to be continuous at $x = a$.
 - 2 Determine whether $f(x) = \frac{x^2 - 4}{x - 2}$ (for $x \neq 2$, with $f(2) = 4$ defined separately) is continuous at $x = 2$.
-

Finding values that ensure continuity

- 3 Find k so that $f(x) = kx^2$ for $x \leq 2$, and $f(x) = 5x - 2$ for $x > 2$, is continuous at $x = 2$.
 - 4 Find a and b so that $f(x) = 2$ for $x \leq -1$, $f(x) = ax + b$ for $-1 < x < 2$, and $f(x) = 6$ for $x \geq 2$, is continuous everywhere.
-

Types of discontinuity

- 5 Classify the discontinuity of each function at the indicated point:
 - a $f(x) = \frac{x+3}{x^2-9}$ at $x = -3$
 - b $f(x) = \lfloor x \rfloor$ at $x = 2$
 - c $f(x) = \frac{1}{(x-1)^2}$ at $x = 1$
-

The Intermediate Value Theorem

- 6 Show that $f(x) = x^3 - 3x + 1$ has a root in $(0, 1)$, using the IVT.
 - 7 A continuous function g satisfies $g(1) = -3$ and $g(5) = 7$. Explain why g must take the value 0 somewhere in $[1, 5]$, and state the theorem used.
 - 8 A cup of coffee's temperature (in $^{\circ}\text{C}$) is a continuous function of time, starting at 90°C and cooling to 25°C over 30 minutes. Explain, using the IVT, why the coffee must have been exactly 50°C at some point.
-

Continuity on an interval

- 9 Determine the largest interval(s) on which $f(x) = \sqrt{9 - x^2}$ is continuous.

- 10 Determine all values of x where $f(x) = \frac{x+1}{x^2-x-6}$ is discontinuous, and classify each discontinuity.
-

Using the IVT to approximate roots

- 11 Show that $f(x) = \cos x - x$ has a root in $(0, 1)$, then use the IVT with the midpoint to narrow the interval to a length-0.5 subinterval.